Exe of Chap 1.

1. Prove that the Cantor set C is totally disconnected and perfect

proof: (1) $\forall x<y\in C$

$\exists k>0$ s.t. $|x-y|>\frac{1}{3^k}$. then x, y will be seperated by two intervals

(2) Given $x\in C$. $\forall k>0$ $\exists$ end-point y_k. s.t. $|x-y_k| \le$ length of $C_k = \frac{1}{3^k}$ $\square$

2. (a) Every number in $[0,1]$ has a ternary expansion:

$x=\sum_{k=1}^{\infty} a_k\cdot 3^{-k}$ $a_k = 0, 1$ or 2. Prove that $x\in C$ iff x has a representation with $a_k = 0$ or 2.

(b). The Cantor-Lebesgue function $F: C\to[0,1]$ $\sum_{k=1}^{\infty} a_k\cdot 3^{-k} \longmapsto \sum_{k=1}^{\infty} b_k\cdot 2^{-k}$. $b_k=\frac{a_k}{2}$.

where $a_k\in\{0,2\}$. Show that F is well-defined, and continuous $F(0)=0$, $F(1)=1$

(c). Prove that F is surjective.

(d). Extend F to be a function continuously on $[0,1]$.

Proof: (a). Consider $F_0: C\to\{0,2\}^{\mathbb{N}}$ If x belongs to $[0,\frac{1}{3}]$ then $\pi_1\circ F_0(x)=0$

If x belongs to $[\frac{2}{9},\frac{1}{3}]$ then $\pi_2\circ F_0(x)=2$

...

On every sheaf, we consider whether x belongs to the left hand or the right hand

$\Rightarrow$ the following number will be 0 or 2 $\Rightarrow a_k\in\{0,2\}$.

(b). 1°. Well-defined: Suppose that $x=\sum_{k=1}^{\infty} a_k\cdot 3^{-k} = \sum_{k=1}^{\infty} a_k'\cdot 3^{-k}$ $a_k, a_k'\in\{0,2\}$.

The minimal number k s.t $a_k\neq a_k'$ let $a_k=0$ $a_k'=2$

Then $\sum_{m\ge k} a_m\cdot 3^{-m} = \sum_{m\ge k+1} a_m\cdot 3^{-m} \le \frac{1}{3^{k+1}} < \frac{2}{3^k} < \sum_{m\ge k} a_m'\cdot 3^{-k}$

contradiction!

2°. Continuouty: $\forall x_0\in C$. $:=\sum_{k=1}^{\infty} a_{k_0}\cdot 3^{-k}$. $\forall\varepsilon>0$ $\exists N>0$ s.t. $\sum_{k\ge N}\frac{1}{2^{N+1}}<\varepsilon$

$\forall\ |x-x_0|<\frac{1}{3^{N+1}}$ we have $a_1=a_{1_0}, \dots, a_N=a_{N_0}$

$\Rightarrow b_1=b_{1_0}, \dots, b_N=b_{N_0} \Rightarrow |F(x)-F(x_0)| < \sum_{k\ge N+1}\frac{1}{2^{N+1}}<\varepsilon$

Thus F is continuous

3°. $F(0)=0$, $F(1)=1$ is trivial.

(c). $\forall y\in[0,1]$: $=\sum_{k=1}^{\infty} b_k\cdot 2^{-k}$ $b_k\in\{0,1\}$ let $x=\sum_{k=1}^{\infty} 2b_k\cdot 3^{-k}$. we have $\begin{cases} F(x)=y \\ x\in C. \end{cases}$ $\square$

(d). We can make induction to show that $\forall k>0$, the new end-points of C_k share the same value of $\frac{1}{2^k}$.

$k=1$, $F(\frac{1}{3})=F(\frac{2}{3})=\frac{1}{2}$

Assume k. consider $k+1$, the new end-point of C_{k+1}: $x_{k+1}=x_k+\frac{1}{3^{k+1}} \Rightarrow F(x_{k+1})=\frac{1}{2^{k+1}}$

Then let all intervals that are out of $C \longmapsto$ the end-point of C $\square$

3.4. (skipped)

5. $E \subseteq \mathbb{R}^d$ O_n open with $O_n = \{x \mid d(x,E) < \frac{1}{n}\}$. Show that:

(a) If E is compact, then $m(E) = \lim_{n\to\infty} m(O_n)$.

(b) When E closed and unbounded / open and bounded, show the failed example.

Proof: (a) $E = \bigcap_{n\geq 1} O_n$. (E closed $\Rightarrow$ $d(x,E)=0$ iff $x \in E$) $\Rightarrow O_n \searrow E$

$m(E) < \infty \implies m(E) = \lim_{n\to\infty} m(O_n)$

(b). $E =$

6. Using translations and dilations, prove the following: Let B be a ball in $\mathbb{R}^d$ of radius r.

Then $m(B) = v_d \cdot r^d$. where $v_d = m(B_1)$, $B_1 = \{x \in \mathbb{R}^d \mid \|x\| < 1\}$.

proof: It's obvious. $\square$

7. If $\delta = (\delta_1, \dots, \delta_d)$ is a d-tuple of $\delta_i > 0$ and E is a subset of $\mathbb{R}^d$.

We define $\delta E := \{(\delta_1 x_1, \dots, \delta_d x_d) \mid (x_1, \dots, x_d) \in E\}$.

Prove that δE is measurable iff E is measurable and $m(\delta E) = \delta_1 \cdot \dots \cdot \delta_d \, m(E)$.

proof: $f : \mathbb{R}^d \to \mathbb{R}^d$ $(x_1, \dots, x_n) \longmapsto (\delta_1 x_1, \dots, \delta_n x_n)$ is a homeomorphism

$\forall \varepsilon > 0 \cdot \exists$ Q_j ~~almost~~ disjoint (almost) cubes (closed) s.t. $m_*(\bigcup_{j=1}^{\infty} Q_j - E) < \varepsilon$

$\Rightarrow \exists$ P_j almost disjoint closed cubes s.t $\bigcup_{j=1}^{\infty} Q_j - E \subseteq \bigcup_{j=1}^{\infty} P_j$

$\Rightarrow m_*(\delta(\bigcup_{j=1}^{\infty} Q_j) - \delta E) \leq m_*(\delta \bigcup_{j=1}^{\infty} P_j) < \delta_1 \dots \delta_n \varepsilon \Rightarrow \delta E$ is measurable

$m(E) \leq m(\bigcup_{j=1}^{\infty} Q_j - E) + m(E) = m(\bigcup_{j=1}^{\infty} Q_j) \leq m(E) + \varepsilon$

$\Rightarrow m(\delta E) \leq m(\bigcup_{j=1}^{\infty} \delta Q_j - \delta E) + m(\delta E) = m(\bigcup_{j\geq 1} \delta Q_j) \leq m(\delta E) + \underset{\delta_1 \dots \delta_n \varepsilon}{\ell} \Rightarrow m(\delta E) = \delta_1, \dots, \delta_d \, m(E)$ $\square$

$\| \; \delta m(\bigcup_{j\geq 1} Q_j)$

4. Cantor-like sets.

Constract a closed set $\hat{C}$ so that at the kth stage of the construction, one removes 2^{k-1} centrally situated open intervals each of length, with $l_1 + 2l_2 + \dots + 2^{k-1}l_k < 1$

(a). If l_j are chosen small enough, s.t. $\sum_{k=1}^{\infty} 2^{k-1}l_k < 1$. Then show that $m(\hat{C}) > 0$

(b) Show that if $x \in \hat{C}$, then $\exists \{x_n\}$ s.t. $x_n \notin \hat{C}$, yet $x_n \to x$, $x_n \in I_n$, where I_n is a sub-interval in the complement of $\hat{C}$ with $|I_n| \to 0$

(c) Prove as a consequence: $\hat{C}$ is perfect, and contains no open interval

(d) $\hat{C}$ is uncountable.

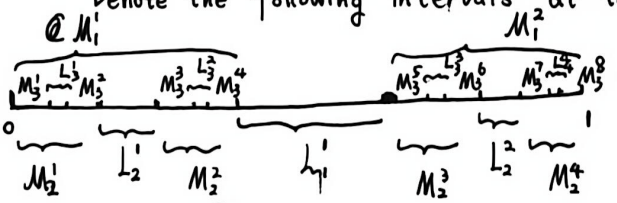
Proof: (a). Denote the ~~denot~~ delected intervals at the kth stage: $L_k^1, \dots, L_k^{2^{k-1}}$

Then $\hat{C} = \bigcap_{k=1}^{\infty} [0,1] \setminus (L_k^1 \cup \dots \cup L_k^{2^{k-1}})$. is a closed set. $\Rightarrow$ $\hat{C}$ is measurable.

$$\Rightarrow m(\hat{C}) = 1 - m\left(\bigcup_{k=1}^{\infty}(L_k^1 \cup \dots \cup L_k^{2^{k-1}})\right) = 1 - \sum_{k=1}^{\infty} 2^{k-1}l_k.$$

(b). Obviously, $l_k \to 0$ as k tends to infinity.

Denote the following intervals at the kth stage; $M_k^1, \dots, M_k^{2^k}$



Given $x \in \hat{C}$ we construct the desired sequence

Then $\hat{C} = \bigcap_{k=1}^{\infty}(M_k^1 \cup \dots \cup M_k^{2^k})$ $\forall k \geq 1$. $\exists\, 1 \leq j \leq 2^k$. s.t. $x \in M_k^j$.

~~Choose~~ $\exists L_{k+1}^i \subseteq M_k^j$. select $x_k \in L_{k+1}^i$. then $x_k \notin \hat{C}$. $|x_k - x| < |M_k^j|$

$$= \left(1 - \sum_{r=1}^{k-1} 2^{r-1}l_r\right)\frac{1}{2^k}$$

Then we have a sequence $\{x_n\}_{n \geq 2}$. $\begin{cases} |x_n - x| \to 0 \quad (n \to \infty) \\ x_n \notin \hat{C}. \end{cases}$

$x_n \in I_n$. with $|I_n| \to 0$

(c) (i) Perfectness we have proved in (b).

(ii) distonnectedness is obvious

(d) There is a homeomorphism between $\hat{C}$ and C $\square$.

9. Give an example of an open set O, such that $\partial(\bar{O})$ has the positive measure

Proof: Choose $O = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{2^{k-1}} L_k^j$ then $\bar{O} = [0,1]$

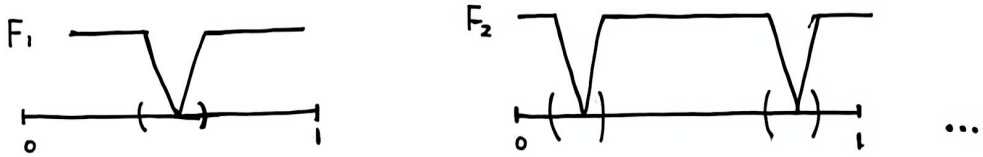
$\Rightarrow m(\partial\bar{O}) = 1 - \sum_{k=1}^{\infty} 2^{k-1} l_k > 0$ □

10. Construct a decreasing sequence of positive continuous functions on $[0,1]$, whose limit is not Riemann integrable.

Proof: Let $\hat{C}$ be the Cantor-like set. $F_1(x) = \begin{cases} 1 & x \in [0,1] \setminus L_1^1 \\ \text{a broken line that equals 0 at the center} & x \in L_1^1 \end{cases}$

$F_2(x) = \begin{cases} 1 & x \in [0,1] \setminus (L_2^1 \cup L_2^2) \\ \text{a broken lines that equals 0 at the center of the intervals} & x \in L_2^1 \\ \cdots & x \in L_2^2 \end{cases}$

let $f_n = F_1 \cdot F_2 \cdot \ldots \cdot F_n$



Then $0 \le f_n(x) \le 1$, $f_n(x) \ge f_{n+1}(x) \Rightarrow \lim_{n\to\infty} f_n(x)$ exists: $= f(x)$

However: f is discontinuous on $\hat{C}$. Since $\forall x \in \hat{C}$. $f(x) = 1$

By the prop in 4, we can construct a sequence $\{x_n\}_{n\ge1}$. $x_n \notin \hat{C}$. $x_n \to x$

W.L.O.G. x_n is selected at the center of the interval I_n $\Rightarrow f(x_n) = 0$

Contradiction!

Thus f has uncountable discontinuouties $\Rightarrow$ f isn't Riemann integrable □.

8. Suppose L is a linear transformation of $\mathbb{R}^d$. Show that if E is measurable, then so is $L(E)$:

(a). Note that if E is compact, so is $L(E)$

(b). Prove the general case

Proof: (a) $L: E \to L(E)$. $x \mapsto Lx$ is a continuous map.

Thus E compact $\Rightarrow L(E)$ compact $\Rightarrow L(E)$ is measurable

(b). $|L(x) - L(x')| \leq M|x - x'|$ M is a constant $\Rightarrow L$ is Lipschitz continuous

$\Rightarrow \forall$ cube P. $\Rightarrow m(L(P)) \leq N\, m(P)$, N is a constant

If $m(E)=0 \Rightarrow \forall \varepsilon>0$, $\exists$ cube P_j s.t. $\sum_{j=1}^{\infty} |P_j| < \varepsilon$

$\Rightarrow m_*(L(E)) \leq \sum_{j=1}^{\infty} N|P_j| < N\varepsilon \Rightarrow m(L(E)) = 0$

Assume E is measurable $\exists$ Borel set F. and G measure zero

s.t. $E = F \cup G$ L is a homeomorphism $\Rightarrow L(E) = L(F) \cup L(G)$ $\begin{cases} L(F) \text{ is a Borel set} \\ m(L(G))=0 \end{cases}$

Thus $L(E)$ is measurable $\square$ (Rmk, If $|L|=0$ then $\dim \mathrm{Im}(L) \leq d-1$. $\Rightarrow m(\dim \mathrm{Im}(L))=0$

9. Find an open set O, s.t. $\partial(\bar{O})$ has a positive measure.

Proof:

Lemma (Cantor-like set): Construct a closed set $\hat{C}$ so that the kth stage of

11. Let A be the subset of $[0,1]$ which consists of all numbers which do not have the digit 4 appearing in their decimal expansion. Find $m(A)$

Solution: Divide $[0,1]$ into 10 pieces. and delete $(\frac{4}{10}, \frac{5}{10})$

Then delete $(\frac{1}{10}\times\frac{4}{10}, \frac{1}{10}\times\frac{5}{10})$, $(\frac{1}{10}+\frac{1}{10}\times\frac{4}{10}, \frac{1}{10}+\frac{1}{10}\times\frac{5}{10})$, ..., $(\frac{9}{10}+\frac{1}{10}\times\frac{4}{10}, \frac{9}{10}+\frac{1}{10}\times\frac{5}{10})$

Keep deleting these intervals and we'll finally get A.

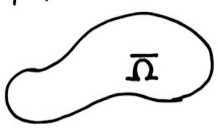
Thus $m(A) = 1 - \frac{1}{10} - 9\times\frac{1}{10^2} - 9^2\times\frac{1}{10^3} - \dots = 1 - \sum_{k=0}^{\infty}\frac{9^k}{10^{k+1}} = 0$ □

2. Prove the following: (a) an open disc in $\mathbb{R}^2$ is not the disjoint union of open rectangles.

(b) An open connected set Ω is the disjoint union of open rectangles iff Ω is itself an open rectangle.

Only prove (b).

proof: (b)



let Ω be an open connected set

Assume that $\Omega = \bigcup_{\alpha\in\Lambda} P_\alpha$. P_α are disjoint open rectangles

Given $\alpha\in\Lambda$, $\forall p\in\partial P_\alpha$. If $\exists\ \beta\in\Lambda$ s.t. $p\in P_\beta$. then by the geometry we have $P_\alpha\cap P_\beta\neq\phi$! thus $\partial P_\alpha\cap\Omega=\phi$.

If $|\Lambda|\geq 2$, Select $x_0\in P_\alpha$. $y_0\in P_\beta$ By the connectedness $\exists$ path $\varphi:[0,1]\to\Omega$ s.t. $\varphi(0)=x_0$, $\varphi(1)=y_0 \Rightarrow \exists\ t\in(0,1)$ s.t. $\varphi(t)\in P_\alpha \Rightarrow P_\alpha\cap\Omega\neq\phi$ Contradiction!

Thus Ω is an open rectangle □.

13. (a) Show that a closed set is a G_δ and an open set is a F_σ

(b). Give an example of an F_σ which is not a G_δ.

(c) Give an example of a Borel set which is not a G_δ nor an F_σ.

Proof: (a) E is a closed set. let $E_n = \{x \mid d(x,E)<\frac{1}{n}\}$ be open

Then $E = \bigcap_{n=1}^{\infty} E_n$ is a G_δ. ⇒

(b) $\mathbb{Q} = \bigcup_{q\in\mathbb{Q}}\{q\}$ is an F_σ but it isn't a G_δ.

(c) $\mathbb{Q}\cup(-1,1)$ is a Borel set □.

14. The outer Jordan content $J_*(E)$ $(E\subseteq\mathbb{R}) = \inf\sum_{j=1}^{N}|I_j|$, $E\subseteq\bigcup_{j=1}^{N} I_j$ by intervals I_j.

(a). Prove that $J_*(E) = J_*(\bar{E})$ $\forall\ E\subseteq\mathbb{R}$.

(b) Exhibit a countable subset $E\subseteq[0,1]$ s.t. $J_*(E)=1$. while $m_*(E)=0$.

Proof: (a). $E\subseteq\bigcup_{j=1}^{N} I_j \Rightarrow \bar{E}\subseteq\bigcup_{j=1}^{N}\overline{I_j}. \Rightarrow J_*(E) = J_*(\bar{E})$

(b) $J_*(\mathbb{Q}\cap[0,1]) = 1$. Indeed, suppose that $\mathbb{Q}\cap[0,1]\subseteq\sum_{j=1}^{N} I_j$

$\Rightarrow [0,1]\subseteq\sum_{j=1}^{N} I_j$ Otherwise. $\exists\ (a,b)\subseteq[0,1]$ s.t. $(a,b)\cap\sum_{j=1}^{N} I_j=\phi$

Contradiction! Thus $J_k(E) = 1$ However $m^*(E) = 0$ $\square$

16. (The Borol – Cantelli lemma). Suppose $\{E_k\}_{k\geq 1}$ measurable $\subseteq \mathbb{R}^d$

and that $\sum_{k=1}^{\infty} m(E_k) < \infty$ Let $E = \{x \mid x \in E_k$ for infinitely many $k\} = \limsup_{n\to\infty} (E_k)$

(a) Show that E is measurable

(b) $m(E) = 0$.

Proof: (a). $E = \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} E_k$ is measurable ($\bigcup_{k\geq n} E_k \subseteq \bigcup_{k\geq 1} E_k$ is measurable)

(b) Let $F_n = \bigcup_{k\geq n} E_k$ then $F_n \supseteq F_{n+1}$, $F_n \searrow E$

Then we have $m(E) = \lim_{n\to\infty} m(F_n) = 0$